Sorting of Data

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Sorting means arrangement of data either in Ascending or descending order.

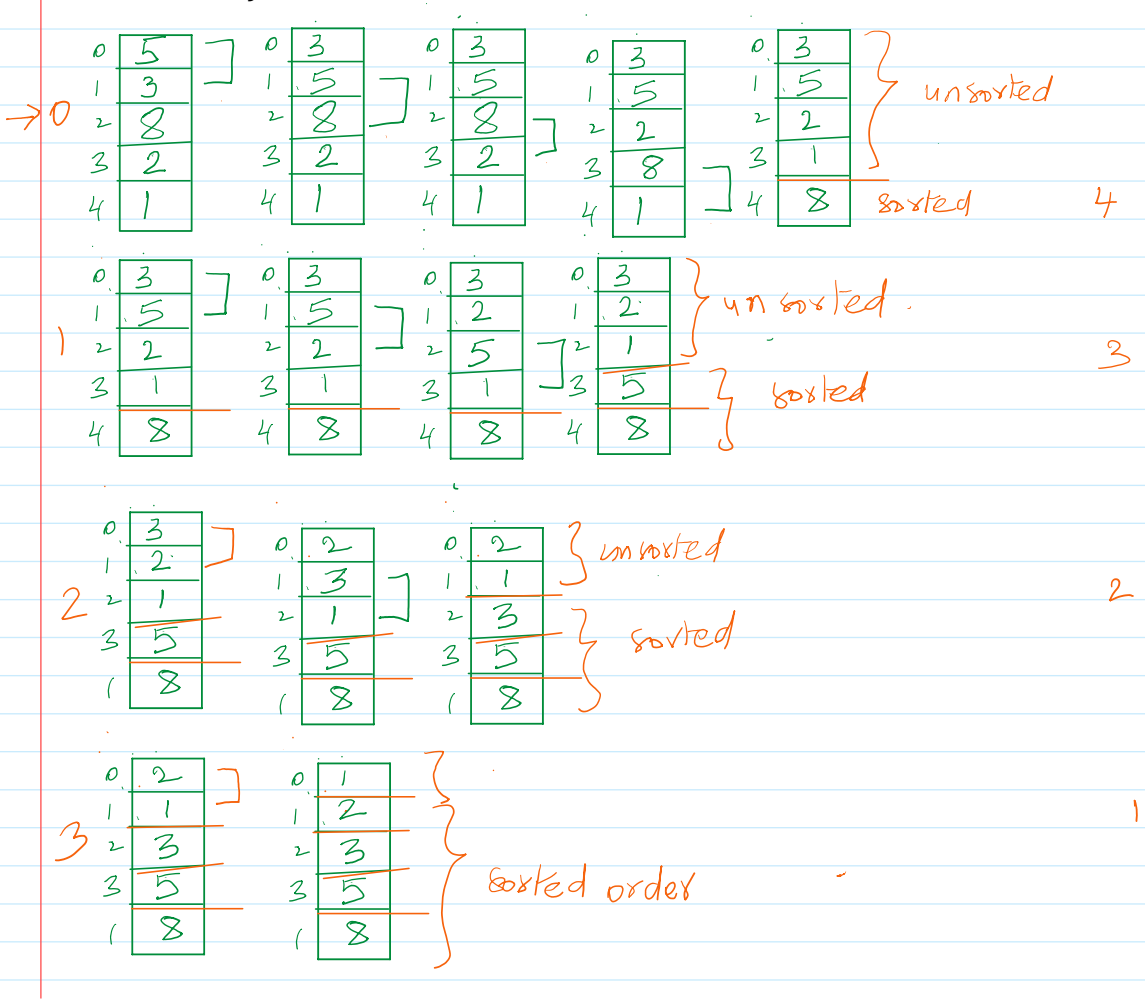
Fundamentally there are 3-types of Sorting.
1. Bubble Sort
2. Selection Sort
3. Insertion Sort

Other Sorting techniques(Advanced Sorting techniques)
i. Merge Sort(Divide and Conquer technique)
ii. Quick Sort(Divide and Conquer technique)

Bubble Sort

What is Bubble Sort?
Bubble sort is a simple sorting algorithm used to arrange elements in ascending or descending order.
It works by repeatedly comparing two adjacent elements and swapping them if they are in the wrong order.
It is called bubble salt because the largest element gradually moves (bubbles) towards the end of the array after every pass.

For example:
Unsorted array:



Total number of comparison for 5 numbers is $4+3+2+1 = 10$

" " " " " 6 numbers is $5 + (4+3+2+1) = 15$

" " " " " 7 numbers is $6 + (5+4+3+2+1) = 21$

Natural number

" " : $(n-1) + (n-2+n-3+\dots 1)$

$= \frac{(n-1)\cdot(n-1+1)}{2}$

$= \frac{n(n-1)}{2}$ comparisons

$= \frac{n^2}{2} - \frac{n}{2}$

$= \frac{1}{2}n^2 - \frac{1}{2}n^1$

constant, n^2 ↑ Major term or Dominating term

What is the sum of natural numbers?

$= \frac{N(N+1)}{2}$ ($N = n-1$)

$1+2+3 = 6$

$1+2+3+4 = \frac{3(3+1)}{2} = 6$

$= \frac{4\times(4+1)}{2} = 10$

In Math or Algorithm analysis there is a function called Big-Oh notation. In this function we pass the major term or dominating term variable along with its exponent. It is represented by Capital 'O'. This function expres the performance of algorithm or code, theoritically. For bubble sort it is $O(n^2)$. Here, 'n' is number of elements or inputs.

By looking at the function, we can say that, if number of elements is increasing, the performance of code is decreasing. Hence, we can say that Bubble sort cannot be used to sort large number of elements.

n	$O(n^2)$
1	1
2	4

1	1
2	4
3	9
4	16
5	25
6	36
...	...
10	100
100	10000
1000	10^6

How does Bubble Sort works?

We compare adjacent elements:
- If the first element is greater than the second, swap them.
- Otherwise, keep them as they are.
- Continue until the end of the array.
- Repeat the process until the array is sorted.

```
package MyArray.Sorting;
import java.util.*;


/**
 * Write a description of class BubbleSort here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class BubbleSort
{
    public void bubbleSort(int A[]){
        int i, j;
        for(i = 0; i < A.length - 1; i++){
            for(j = 0; j < A.length - 1 - i; j++){
                if(A[j] > A[j+1]){
                    int t = A[j];
                    A[j] = A[j+1];
                    A[j+1] = t;
                }
            }
        }
    }

    public void printArray(int A[], String msg){
        System.out.print("\n" + msg + "\n[");
        int i;
        for(i = 0; i < A.length - 1; i++){
            System.out.print(A[i]+", ");
```

quadratic time

$O(n^2)$

H.W : If sorted array is given to this function, it is still taking $O(n^2)$, your job is, if sorted array is given, change the code in such a way that it will take $O(n)$ comparison.

linear time

```
        }
        System.out.print(A[i]+"]\n");
    }

    public int[] createRandomArray(int size){
        int A[] = new int[size];
        Random r = new Random();
        int i;
        for(i = 0; i < size; i++){
            A[i] = r.nextInt(10, 100);
        }
        return A;
    }
}
```

Output:

